



Ultrapure single-band red upconversion luminescence in Er^{3+} doped sensitizer-rich ytterbium oxide transparent ceramics for solid-state lighting and temperature sensing

CONG ZHANG,^{1,2} YANLI SHI,^{1,2,*} KAILEI LU,^{1,2} XIULING WANG,^{1,2}
HAIFENG YUAN,^{1,2} RUICHONG CHEN,³ JIANQI QI,^{1,2}
AND TIECHENG LU^{1,2}

¹College of Physics, Sichuan University, Chengdu 610064, China

²Key Laboratory of Radiation Physics and Technology of the Ministry of Education, Sichuan University, Chengdu 610064, China

³School of Mechanical Engineering, Chengdu University, Chengdu 610106, China

*shiyanni@scu.edu.cn

Abstract: Achieving single-band upconversion (UC) is a challenging but rewarding approach to attain optimal performance in diverse applications. In this paper, we successfully achieved single-band red UC luminescence in Yb_2O_3 : Er transparent ceramics (TCs) through the utilization of a sensitizer-rich design. The Yb_2O_3 host, which has a maximum host lattice occupancy by Yb^{3+} sensitizers, facilitates the utilization of excitation light and enhances energy transfer to activators, resulting in improved UC luminescence. Specifically, by shortening the ionic spacing between sensitizer and activator, the energy back transfer and the cross-relaxation process are promoted, resulting in weakening of green energy level $^4\text{S}_{3/2}$ and $^2\text{H}_{11/2}$ emission and enhancement of red energy level $^4\text{F}_{9/2}$ emission. The prepared Yb_2O_3 : Er TCs exhibited superior optical properties with in-line transmittance over 80% at 600 nm. Notably, in the 980nm-excited UC spectrum, green emission does not appear, thus Yb_2O_3 : Er TCs exhibit ultra-pure single band red emission, with CIE coordinates of (0.72, 0.28) and color purity exceeding 99.9%. To the best of our knowledge, this is the first demonstration of pure red UC luminescence in TCs. Furthermore, the luminescent intensity ratio (LIR) technique was utilized to apply this pure red-emitting TCs for temperature sensing. The absolute sensitivity of Yb_2O_3 : Er TCs was calculated to be 0.319% K^{-1} at 304 K, which is the highest level of optical thermometry based on $^4\text{F}_{9/2}$ levels splitting of Er^{3+} known so far. The integration between pure red UC luminescence and temperature sensing performance opens up new possibilities for the development of multi-functional smart windows.

© 2023 Optica Publishing Group under the terms of the [Optica Open Access Publishing Agreement](#)

1. Introduction

Since discovered by Auzel in 1966, upconversion luminescence (UCL) which is capable of converting near-infrared (NIR) photons into visible emission, has gained considerable attention [1–4]. Due to superior properties such as high chemical stability, excellent photostability, sharp emission band, and weak background autofluorescence, UCL materials in diverse forms have been extensively investigated for various applications, including solid state laser, optical sensors, lighting, solar cell, high-resolution displays, and more [5–9]. In the past decades, Yb^{3+} co-doped with Er^{3+} , Tm^{3+} , and Ho^{3+} etc. systems remain the key constituents for the majority of red, green, and blue UCL material [2,10]. Typically, Yb^{3+} act as the sensitizer in UC process by virtue of the large absorption cross-sections around 980 nm, which match well with commercial GaAs lasers. Er^{3+} is commonly selected as the activator due to its high UC efficiency and emission tuning properties. Due to the abundant excitation energy levels of Er^{3+} , $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped UC materials

generally exhibit multiple emission bands with green emission at ~520-550 nm and red emission at ~660 nm. While some applications benefit from this multi-band UC emission characteristics, such as the realization of white UC light and the acquisition of multicolor UC, other applications, like color display [5], security coding [11], and multiplexed molecular imaging [12], require the utilization of single-band pure UC light for the refinement or determination of the emitted color [13,14]. Additionally, considering that multi-band emissions result in the allocation of available energy into multiple frequencies, single-band emission is preferred when a high conversion efficiency is required in photonic devices, such as in nonlinear photonic devices [5]. Specifically, the NIR region of 650-1000 nm is typically considered the first biological window, as it exhibits weak tissue absorption, light scattering, and autofluorescence [15]. For example, in bio-imaging applications, short wavelength emission bands outside the biological window ($\lambda < 650$ nm) can interfere with the bio-imaging signals, resulting in reduced spatial resolution and sensitivity [16]. The red UC emission of Yb/Er system, located in the first biological window (BW-I), has a stronger penetration ability compared to green UC emission, which effectively minimizes background interference and improves detection efficiency [17]. Hence, tuning both emission and excitation bands of UC into the biological window will improve the anti-interference capabilities of UCL devices to the external environment [18,19].

To attain pure red UCL in Er^{3+} -doped materials, many efforts have been made to design single-band emission UC materials of high purity. One common approach is doping ions with suitable energy levels, such as Mn^{2+} . For example, Tian et al. obtained single-band red UC from $\text{NaYF}_4:\text{Er}^{3+}/\text{Yb}^{3+}$ nanoparticles via Mn^{2+} doping with effective energy transfer (ET) occurring between Mn^{2+} and Er^{3+} ions [20]. Similar results have been achieved using Mn^{2+} -containing hosts such as NaMnF_3 , KMnF_3 , and MnF_2 [21–23]. However, owing to the considerable changes in valence state and ionic radius between the Ln^{3+} and Mn^{2+} ions, undesirable modifications to the size and phase of $\text{NaLnF}_4:\text{Yb/Er}$ were also introduced. Moreover, it was found that Mn^{2+} co-doping weakened the UC intensity in NaYF_4 [24,25]. Another approach is to select a specific matrix, such as KMgF_3 , KSc_2F_7 , $\text{Na}_x\text{ScF}_{3+x}$, ScOF , Na_3ZrF_7 , or Na_3HfF_7 , which allows for the attainment of single-band UCL [26–33]. However, the specific factors, whether structural or electronic, responsible for the single-band emission in these matrices are still uncertain. Other methods, like coating with organic dyes to extinguish green emission, are employed to generate single-band UC, but eliminating green emission entirely is challenging [13,16]. Therefore, to obtain single-band emission without reducing the UC efficiency, modulating the spectral emission by effective ET process such as EBT and CR is a more promising approach [16].

In most cases, altering the sensitizer to activator ratio can adjust red/green emission ratio in the $\text{Yb}^{3+}/\text{Er}^{3+}$ system. The EBT between the sensitizer and the activator inhibits green emission, which can be augmented by increasing the sensitizer concentration. Based on this principle, single-band red UCL can be easily realized in high sensitizer to activator ratio material [34]. Actually, single-band red emissions have been obtained from sensitizer-rich Yb-based materials such as $\text{SrYbInO}_4:\text{Er}$, $\text{YbOF}:\text{Er}$, $\text{YbOCl}:\text{Er}$, and $\text{Yb}_3\text{Al}_5\text{O}_{12}:\text{Er}$ [35–39]. Therefore, a feasible strategy is to improve the probability of EBT process by shortening the distance between sensitizer and activator, reducing the green emission and thus obtaining a pure red UCL [40]. Here, Yb_2O_3 was chosen as the host, which is distinctive as all cations act as sensitizers. Due to the single excitation energy level of Yb^{3+} , it still can mediate effective long-range energy transfer/migration to stimulate activators even at high concentrations when used as a sensitizer [41]. The reduced ionic spacing is predicted to generate strong Yb-Er interactions, potentially resulting in highly efficient UC performance [42]. Additionally, Yb_2O_3 can fully utilize the 980 nm energy flux due to its large absorption cross-section. Moreover, oxides tend to possess greater chemical, thermal, and mechanical stability compared to fluorides, making them appropriate hosts for UCL applications [2,13].

For transparent bulk materials, smart windows are a promising application scenario. Recently, numerous novel types of smart windows have been developed for various applications, such as temperature sensing, switchable sunroof, and optical data recording and storage [43–45]. Conventional transparent materials such as glass or plastics have been considered as multifunctional window materials in cooperation with RE^{3+} ions, but they may be limited by low thermal and mechanical properties in harsh environments [46–48]. A better alternative is transparent ceramic (TC) due to its intriguing geometric versatility, doping flexibility, and manufacturing scalability compared to single crystals [49–51]. Typical transparent ceramics such as YAG, ALON etc. have been widely investigated as solid-state laser, armor, and phosphor materials [52–54]. Yb_2O_3 belongs to the cubic crystal system and does not undergo the birefringence effect, so it has the prerequisite to becoming a transparent ceramic [55]. Therefore, we aim to make Yb_2O_3 powder into fully-dense bulk ceramics with desirable transparency, high thermal conductivity and thermal stability, and excellent mechanical properties [56]. These Yb_2O_3 TCs are expected to exhibit superior spectroscopic properties compared to those of powders due to the reduction of adverse light scattering.

In this work, we prepared a series of highly transparent Er-doped Yb_2O_3 ceramics. The highest transmittance of Yb_2O_3 : Er TC reach 82% at 600 nm, which is the highest value of Yb/Er doped transparent ceramics reported so far. The realization of 980 nm-excited single-band red UCL in TCs was demonstrated. The color purity of red UCL is calculated to be >99.9% and the single-band characteristic is independent of the doping concentration and pump power. The temperature sensing performance of the pure red emission was evaluated using the luminescent intensity ratio (LIR) technique. The Yb_2O_3 : Er TCs possesses the advantages of red UCL's high penetration and anti-interference resistance, as well as the outstanding physical and chemical characteristics of transparent ceramics. Our work provides a new idea for the preparation and application of multi-functional window materials with both lighting and sensing capabilities.

2. Experimental

2.1. Sample preparation

Commercial Yb_2O_3 and Er_2O_3 (99.99%, Xingda, China) powders were chosen as the starting materials. Based on our previous work, ZrO_2 (99.99%, Dahua, China) was employed as sintering additives [55]. On the basis of the ratio of $(\text{Yb}_{0.97-x}\text{Er}_x\text{Zr}_{0.03})_2\text{O}_3$ ($x = 0.01, 0.03, 0.06, 0.1, 0.2$), powders were stoichiometric weighted and mixed with ethanol in a 500 mL nylon jar. The weight ratio of zirconia balls ($\Phi 5$ mm) to raw material was set at 18:1, and the ball milling speed was 250 rpm for 20 h. After mixing, the slurries were dried at 70 °C for 24 hours, then the powder was sieved through a 200-mesh sieve. After calcined at 1200 °C for 4 h, these pretreated powders were dry-pressed into 20 mm diameter and 2 mm thick disks under uniaxial pressure of 20 MPa. Then, the pellets were cold isostatically-pressed at 200 MPa for 10 min. The sintering process was carried out in a vacuum carbon canister furnace at 1800 °C for 10 h under a vacuum of 10^{-3} Pa. Then the samples were annealed at 1450 °C for 20 hours in air to remove oxygen vacancy. Finally, these sintered Yb_2O_3 TCs disks were machined to a thickness of 1.5 mm and polished on both sides for further characterization. The preparation flow chart of Yb_2O_3 : Er TCs is shown in [Supplement 1](#), Fig. S1.

2.2. Measurements and characterization

X-ray Diffraction (XRD, DX-2700, Dandong Fangyuan, China) was used to determine the compositions and phase structures of powders and annealed TCs using nickel-filtered $\text{Cu-K}\alpha$ radiation ($\lambda = 1.5406 \text{ \AA}$), and the diffraction patterns were recorded in the range of 2θ from 20 to 70°. The scanning electron microscope (SEM, JSM-IT500HR, JEOL Japan) was used to obtain the morphology of as-prepared powders and microstructures of sintered ceramics. At least 100

grains were measured using the linear intercept method on Nanomeasure software to determine the mean disc size. An intensification factor of 1.56 was used to obtain the equivalent mean grain size [57]. The distribution of elements in Yb₂O₃: Er TCs was obtained with a Quantax EDS (Energy Dispersive X-ray Spectrometer) detector (Aztec Energy X-Max 20, Oxford, UK). The density of TCs was measured by Archimedes method. The Raman spectrum were used to determine the phonon energy of the Yb₂O₃ matrix using a Raman spectrometer (HR800; Horiba, France) with a 532 nm excitation source. The in-line transmittance of TCs was measured with UV/Vis/NIR spectrophotometer (Lambda-750, Perkin-Elmer, USA). UC fluorescence spectra and lifetime were measured on a Fluorolog-3 spectrofluorometer (Horiba JobinYvon, France) with a 980 nm laser (MDL-III-980) as the excitation source and a picosecond photon detection module (PPD-850) as the detector. The heating apparatus was constructed using an alumina disk regulated by a custom-made control unit. To minimize temperature inaccuracies, two channels of a thermometer were employed for both calibration and temperature measurement.

3. Results and discussion

3.1. Structural and optical properties

XRD patterns of Yb₂O₃: Er TCs and precursor powders are shown in Fig. 1(a) and Supplement 1, Fig. S2, respectively. Clearly, the crystalline phases of all samples are identical to the standard cubic Yb₂O₃ phase (PDF #43-1037). No secondary phase was observed, regardless of the doping concentration, indicating that Er³⁺ ions were successfully incorporated into the Yb₂O₃ lattice, substituting for Yb³⁺. The lattice parameters of the formed Yb₂O₃: Er were refined from XRD patterns to evaluate the impact of the concentration of Er³⁺ on its incorporation into the matrix. The estimated lattice constants from the Bragg equation are 10.434 Å, 10.438 Å, 10.439 Å, 10.444 Å, and 10.451 Å for the specimens 1%, 3%, 6%, 10%, and 20% respectively. The lattice constant linearly increases with the increase of Er³⁺ concentration. This is due to the slightly larger ionic radius of Er³⁺ (0.89 Å, coordination number (CN) = 6) compared to that of Yb³⁺ (0.868 Å, CN = 6).

Figure 1(b) shows the Raman spectrum of undoped Yb₂O₃ TCs. The most prominent band is observed at 362 cm⁻¹, which is the characteristic peak of cubic Yb₂O₃ and it is ascribed to Ag + Fg vibrations [55]. In addition, the maximum phonon energy of Yb₂O₃ was observed to be 614 cm⁻¹, which is lower than commonly used oxide UC hosts such as YAG (850 cm⁻¹) and YPO₄ (1080 cm⁻¹). The low phonon energy host helps to reduce non-radiative processes that can lead to energy loss, which is beneficial for photoluminescence and UC applications. The crystal structure of Yb₂O₃ is depicted in the inset of Fig. 1(b). Yb₂O₃ crystallizes in the cubic space group Ia3(206) with two six-coordinate sites (C₂, C_{3i}). 24 Yb³⁺ are on the sites with point symmetry C₂ and 8 Yb³⁺ ions are on the sites with symmetry C_{3i}. Due to the inversion symmetry of the C_{3i} sites, the optical characteristics are almost exclusively attributed to the ions on C₂ sites [58].

Generally, the density is required to be greater than 99% of the theoretical density for the ceramic to become transparent. Thus, the density of this ceramic is expected to be high because both Yb and Er have large atomic numbers. The density of 1%, 3%, 6%, 10% and 20% Er³⁺ doped Yb₂O₃ transparent ceramic measured by Archimedes method was 8.971 g/cm³, 8.968 g/cm³, 8.927 g/cm³, 8.922 g/cm³, and 8.870 g/cm³ respectively. Since the relative atomic mass of Yb (173.04) is larger than that of Er (167.26), the density of the Yb₂O₃ TCs sample decreases slightly with increasing Er content. It is well known that scintillator and X-ray imaging applications usually require high atomic numbers and densities [59,60]. This means that the Yb₂O₃: Er in the form of transparent ceramic may also have potential applications in X-ray CT imaging and will be reported in our future work.

Figure 1(c) and Supplement 1, Figs. S3-S5 show SEM images and EDS elemental distributions of the (Yb_{0.97-x}Er_xZr_{0.03})₂O₃ powders and thermally etched ceramics. All samples present

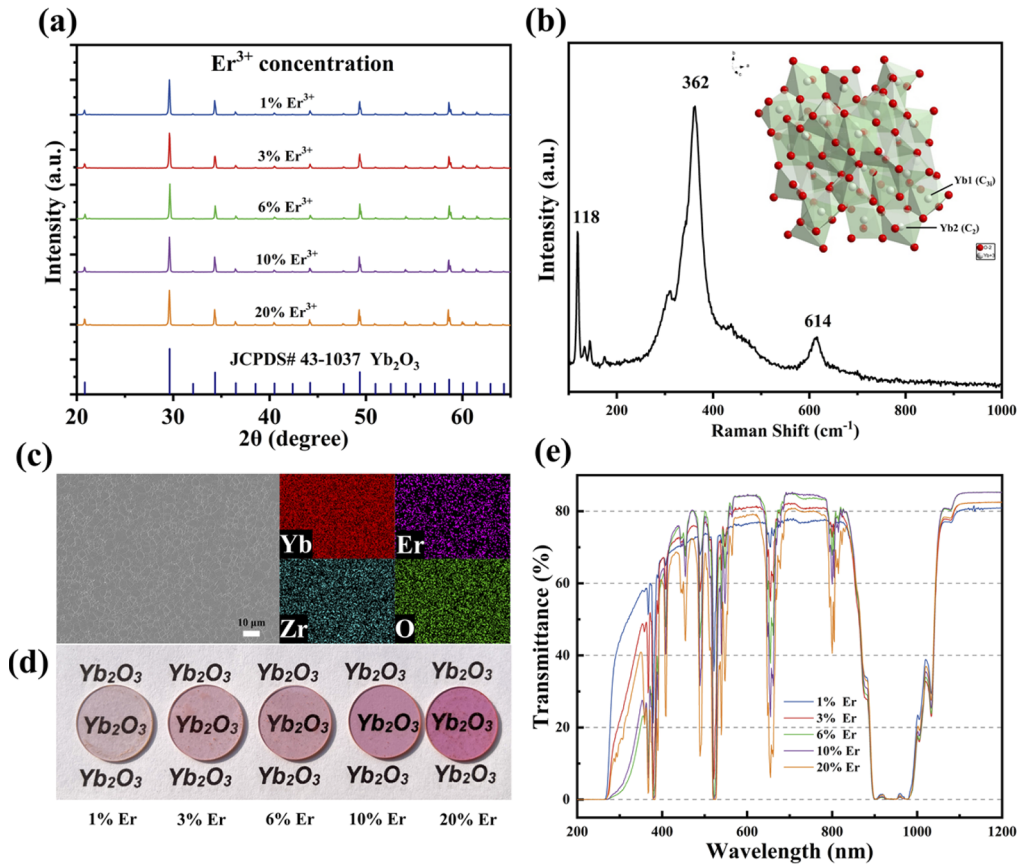


Fig. 1. (a) XRD pattern (b) Raman spectra (c) EDS mapping (d) photograph (e) In-line transmittance of Yb_2O_3 : Er TCs.

homogeneous microstructure without abnormal grain growth. Furthermore, all the grains are firmly bonded together with virtually no pores or impurities present. The average grain size of ceramics obtained from SEM images is approximately $3.5 \mu\text{m}$, regardless of the concentration of Er^{3+} doping. This may be attributed to the similarities in the crystal structure of Er_2O_3 and Yb_2O_3 . According to the Rayleigh approximation, the intensity of scattered light is to the sixth power of grain size, which means that a smaller average size distribution leads to higher transmittance [61]. The elemental distribution on the surface confirms that the Yb, Er as well as Zr elements are uniformly distributed in the powder and ceramic. The dense and uniform microstructure of the ceramics contributes to its high level of transparency.

To better understand the optical properties, photographs and in-line transmittance of the Yb_2O_3 : Er TCs are depicted in Fig. 1(d) and Fig. 1(e). The letters under the TCs are clearly visible. These TCs samples appear pink to the naked eye because of the large absorption of red and green light in visible region, and the pink deepens with increasing Er^{3+} content. These absorption bands in the transmission spectra originate from the electron transitions within the $4f^{12}$ subshell of Er^{3+} . All samples show high transparency in part of visible and infrared regions, with the transmittance of $\sim 82\%$ at $\sim 600 \text{ nm}$. The difference in transmittance is primarily attributed to the presence of porosity and impurities introduced during the preparation process, as there is no absorption of Yb and Er ions at 600 nm . Supplement 1, Table S1 states the transmittance of Yb/Er doped TCs reported so far. It is worth mentioning that the transmittance of the Yb_2O_3 : Er TCs in

this work is the highest level of Yb-Er doped TCs that have been reported. This indicates that even with such high doping concentrations, the prepared samples still have high optical quality. The absorption bands in the range from 900 to 1000 nm are ascribed to $^2F_{7/2} \rightarrow ^2F_{5/2}$ electronic transitions of Yb^{3+} ions [62]. Transmittance of all samples was almost 0 in this range owing to the high Yb^{3+} concentration. The absorption spectra can be derived from transmittance spectra by the formula: $\alpha = 1/d \cdot \ln(1/T)$, where α is the absorption coefficient, d is the thickness of samples, and T is the transmittance. The calculated absorption spectra were shown in Supplement 1, Fig. S6. These absorption peaks correspond to $\text{Yb}^{3+}: ^2F_{5/2}$ and $\text{Er}^{3+}: ^4S_{3/2}, ^4F_{9/2}, ^4I_{9/2}, ^4I_{11/2}, ^4I_{13/2}$, respectively, and detailed schematic diagram of the energy level transitions can be found in Supplement 1, Fig. S7 [63]. The intensity of absorption peaks from Er^{3+} increases with the increase of Er^{3+} concentration. The strong and broad absorption band in the 900-1000 nm range is mainly attributed to the Yb_2O_3 matrix, corresponding to $^2F_{7/2} \rightarrow ^2F_{5/2}$ transitions of Yb^{3+} as well as $^4I_{15/2} \rightarrow ^4I_{11/2}$ transitions of Er^{3+} ions. As a result, it has been proven that the sample is of good quality and has a relatively high transmittance, making it suitable for luminescence and sensing tests.

3.2. UCL properties

The UCL spectra of Yb_2O_3 TCs doped with different Er^{3+} concentrations under 980 nm-laser excitation were measured as demonstrated in Fig. 2(a). Several emission peaks are observed in the red region with maximum intensity at 662 nm, originating from the $^4F_{9/2} \rightarrow ^4I_{15/2}$ transition of Er^{3+} . Generally, the $^2H_{11/2}$ and $^4S_{3/2} \rightarrow ^4I_{15/2}$ transitions of Er^{3+} also generate green emission. It is surprising to find that green UC emission does not appear at all in the spectrum, leaving a very pure red UC emission. Inset of Fig. 2(a) shows the relationship between the integrated red emission intensity and the concentration of Er^{3+} doping. The UCL intensity gradually becomes stronger at first and decreases when the Er^{3+} concentration exceeds 6%. There are two possible mechanisms to explain this phenomenon. Firstly, this phenomenon could be attributed to non-radiative relaxation processes, such as cross-relaxation, which become more prominent with an increase in the concentration of Er^{3+} ions [64]. Secondly, the notable decrease in peak intensity may result from the self-absorption of emitted photons by the Er^{3+} ions themselves. This effect is also evident in the absorption spectra (Fig. S6), where an increase in Er^{3+} concentration leads to higher absorbance. Boccolini et al. found that quantum yield values are constrained by self-absorption losses [6]. Their findings indicate the presence of an optimal thickness and doping concentration of ions involved in the up-conversion/down-conversion process to minimize self-absorption and enhance the photoluminescence quantum yield (PLQY) of the material [65]. The similar phenomenon appears in the downshifting emission spectrum excited by 980 nm as shown in Supplement 1, Fig. S7. The broad emission band around 1530 nm originates from the $^4I_{13/2} \rightarrow ^4I_{15/2}$ transition of Er^{3+} .

Based on the recorded UCL emission spectra, the Commission International de l'Eclairage (CIE) chromaticity of the samples was evaluated, as shown in Fig. 2(b). Color purity is an important indicator to evaluate chromatic characteristics of luminescent materials, which could be calculated using the following formula:[66]

$$\text{Color Purity} = \frac{\sqrt{(x-x_i)^2 + (y-y_i)^2}}{\sqrt{(x_d-x_i)^2 + (y_d-y_i)^2}} \times 100\% \quad (1)$$

(x, y) is the CIE color coordination of $\text{Yb}_2\text{O}_3: \text{Er}^{3+}$ TCs under 980 nm excitation. (x_d, y_d) represents the coordinate corresponding to the dominant wavelength of the light source, and (x_i, y_i) represents the white light source with CIE color coordinate (0.3333, 0.3333). The chromaticity coordinates and calculated color purity were shown in Table 1. The highest color purity of samples was calculated to be ~99.9%, suggesting potential applications as red TC phosphors in

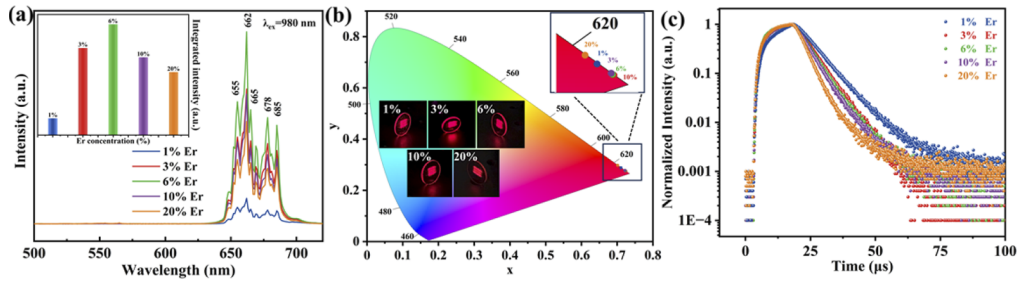


Fig. 2. (a) The UCL spectra of $\text{Yb}_2\text{O}_3:\text{Er}$ TCs with different Er^{3+} concentrations. The inset displays the changes in the UCL intensities in relation to the Er^{3+} concentration. (b) CIE chromaticity diagram of the $\text{Yb}_2\text{O}_3:\text{Er}^{3+}$ TCs. The inset presents the photograph of the TCs excited by 980 nm laser diode. (c) fluorescence decay curves of emission at 662 nm.

lighting and display devices [67]. The inset of Fig. 2(b) presents the digital photograph of the $\text{Yb}_2\text{O}_3:\text{Er}^{3+}$ TCs under 980 nm excitation, where the bright red light could be seen clearly.

Table 1. CIE Chromaticity Coordinates of $\text{Yb}_2\text{O}_3:\text{Er}$ transparent ceramics

Sample	CIE chromaticity coordinates		Color purity
	x	y	
$\text{NaErF}_4:\text{Mn}, \text{Yb}$ [86]	0.7181	0.2741	97.66%
1%	0.71065	0.2881	99.63%
3%	0.72371	0.27596	99.73%
6%	0.72267	0.27697	99.70%
10%	0.72153	0.2781	99.90%
20%	0.70123	0.29777	99.72%

To elucidate the luminescence dynamic involved, luminescence decay curves at room temperature of the ceramics were investigated. Figure 2(c) depicts the fluorescence lifetime curves of the $\text{Yb}_2\text{O}_3:\text{Er}^{3+}$ TCs with different Er^{3+} ion concentrations ($\lambda_{\text{ex}} = 980 \text{ nm}$, $\lambda_{\text{em}} = 662 \text{ nm}$). The detected decay curves can be well fitted with the single exponential decay model as follows:

$$I(t) = I_0 + A \exp(-t/\tau) \quad (2)$$

where $I(t)$ and I_0 represent the emission intensity at time t and $t = 0$, respectively, τ represents the lifetime, and A is constant. The lifetimes of $\text{Yb}_2\text{O}_3:\text{Er}$ TCs were calculated to be 5.94, 5.56, 5.31, 4.96, and 4.11 μs corresponding to 1%, 3%, 6%, 10%, and 20% Er^{3+} concentrations, respectively. The value of lifetime decreases with the increasing Er^{3+} concentration, which further verifies the existence of concentration quenching in $\text{Yb}_2\text{O}_3:\text{Er}$ TCs.

3.3. Excitation mechanism of UCL

To understand the UCL mechanism of the red emission, the dependence of UCL intensity on the pump power of the 980 nm laser was measured as shown in Fig. 3(a). For the unsaturated UCL process, the intensity I and pump laser power P generally abide by the following equation:[68]

$$I \propto P^n \quad (3)$$

where n denotes the number of infrared photons absorbed for each visible photon emitted, which is equivalent to the slope of the line obtained from fitting the $\log I$ versus $\log P$. Clearly, the

intensity of UCL increases with the increase of pumping power. The slopes n of 1%, 3%, 6%, 10%, and 20% for the red emission are calculated to be 0.38, 0.6, 1.53, 1.62, and 1.96, respectively. The relationship between n and Er^{3+} concentration is shown in Fig. 3(b). Surprisingly, the slopes of 1% and 3% Er^{3+} doped sample is 0.38 and 0.6 respectively, which is far deviated from 2 for the two-photon process. The reason may be as follows: The low doping level induces a long distance between the participating activator ions in Yb_2O_3 host material, which is adverse to the ET process but facilitate the EBT to another Yb^{3+} [69]. In this case, fewer photons participate in UC process, resulting in a lower n . The same phenomenon happens in low concentration Ho^{3+} doped Yb_2O_3 nanoparticles [70,71]. When the concentration of Er^{3+} increases, ET from Yb^{3+} to Er^{3+} will result in a more efficient UC from the intermediate energy level to the UCL energy level, leading to an increase in the proportion of UC processes. This could explain the increase of slope n with increasing Er^{3+} concentration.

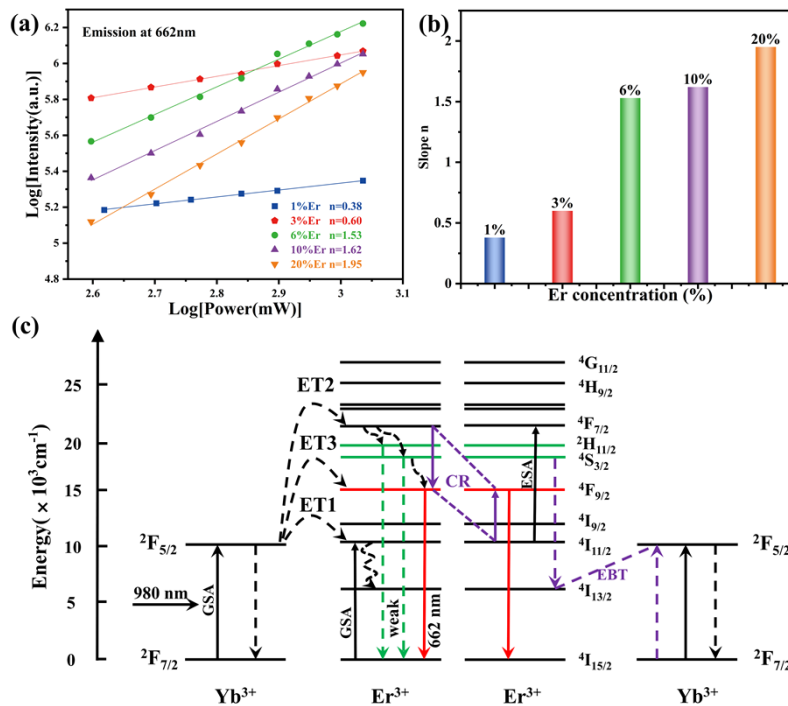
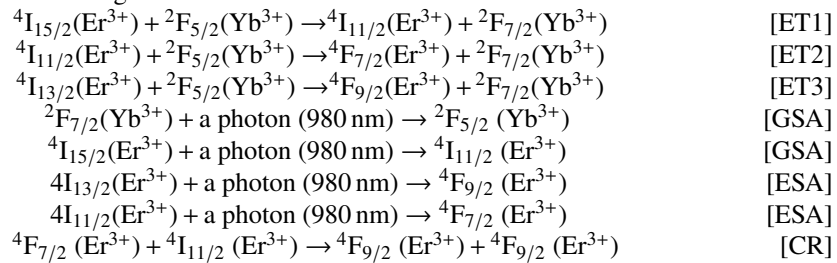
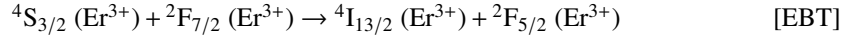


Fig. 3. (a) The pump power dependence of the UCL intensities in the Yb_2O_3 : Er TCs. (b) Relationship between n and Er^{3+} concentration (c) Energy level diagrams of Yb^{3+} , Er^{3+} and possible transition pathways and ET processes upon 980 nm-excitation.

The possible population processes and feasible transitions were schematically given in the energy level diagram of Er^{3+} and Yb^{3+} ions, as depicted in Fig. 3(c). The major UC mechanisms can be depicted as following:





UC excitation processes involve ground-state absorption (GSA), excited-state absorption (ESA), and energy transfer (ET), additional process such as cross relaxation (CR) between two activator ions may occur as well. The GSA or ESA process generally involves in a single ion and it is the dominant UC excitation process when doping level is low, while the ET process involves two adjacent ions and it is dominant in high-doping concentration materials [72]. The ET process from Yb^{3+} to Er^{3+} makes prominent contributions to the population of the emission energy levels because Yb^{3+} has a higher absorption cross section than Er^{3+} at the infrared region and a high doping concentration. Under the 980 nm excitation, the GSA of Yb^{3+} take places via absorbing an infrared photon: ${}^2F_{7/2} + \text{a (980 nm) photon} \rightarrow {}^2F_{5/2}$ transition. Subsequently, since ${}^4I_{11/2}$ of the Er^{3+} and ${}^2F_{5/2}$ of the Yb^{3+} are syntonics, the ET1 process occurs: ${}^4I_{15/2}(\text{Er}^{3+}) + {}^2F_{5/2}(\text{Yb}^{3+}) \rightarrow {}^4I_{11/2}(\text{Er}^{3+}) + {}^2F_{7/2}(\text{Yb}^{3+})$. For populating the ${}^4F_{7/2}$ level of Er^{3+} , the ESA takes place from the ${}^4I_{11/2}$ level to the ${}^4F_{7/2}$ level by absorbing another 980 nm photon. Another route for populating the ${}^4F_{7/2}$ level is through the ET2 process: ${}^4I_{11/2}(\text{Er}^{3+}) + {}^2F_{5/2}(\text{Yb}^{3+}) \rightarrow {}^4F_{7/2}(\text{Er}^{3+}) + {}^2F_{7/2}(\text{Yb}^{3+})$. There are three routes for the red emission originating in the energy level ${}^4F_{9/2}$ [66]. The first route is the electrons at upper level ${}^4S_{3/2}$ and ${}^2H_{11/2}$ non-radiatively relax to ${}^4F_{9/2}$ level. The second is the CR process, in which ${}^4F_{9/2}$ is populated through ${}^4F_{7/2} \rightarrow {}^4F_{9/2}$ and ${}^4I_{11/2} \rightarrow {}^4F_{9/2}$ through CR transfer. As for the third route, the electrons at ${}^4I_{11/2}(\text{Er}^{3+})$ non-radiatively relax to the ${}^4I_{13/2}(\text{Er}^{3+})$, followed by the ESA process: ${}^4I_{13/2}(\text{Er}^{3+}) + \text{photon} \rightarrow {}^4F_{9/2}(\text{Er}^{3+})$ or the ET2 process: ${}^4I_{13/2}(\text{Er}^{3+}) + {}^2F_{5/2}(\text{Yb}^{3+}) \rightarrow {}^4F_{9/2}(\text{Er}^{3+}) + {}^2F_{7/2}(\text{Yb}^{3+})$. According to Eqs. 4 and 6, the probability of cross-relaxation is not high in samples with low Er concentrations, but the samples still obtain red UCL, indicating that the contribution of this process is not significant. As for the NR process and ET3, Wu. et al. compares the PL spectra excited at 488 nm and the UCL spectra excited at 980 nm of the materials Y_2O_3 , BGZO, $\beta\text{-NaYF}_4$, and $\text{Ba}_3\text{Y}_4\text{O}_9$ doped with 5% Er^{3+} , 20% Yb^{3+} and discusses the contribution of the NR and ET3 processes for the red UCL [58,73]. In the PL spectrum, the Er^{3+} ${}^4F_{7/2}$ state will be directly populated, followed by relaxation to the ${}^2H_{11/2}$ and ${}^4S_{3/2}$ levels under 488 nm excitation. Thus, the red emitting level ${}^4F_{9/2}$ will be populated exclusively from the green level via NR. The results indicate that the R/G ratio of these matrices in the UCL spectrum is greater than that in the PL spectrum, which means that the contribution of NR to the red UCL of all samples is not significant. Therefore, the ET3 process is the dominant process of the red UCL in Yb_2O_3 : Er. Further, there exists the energy back transfer (EBT) process from Er^{3+} to Yb^{3+} (${}^2F_{7/2} + {}^4S_{3/2} - {}^2F_{5/2} + {}^4I_{13/2}$) at high Yb^{3+} concentration, leading to a decrease in the green energy level [74]. The processes discussed above could cause an increase at the ${}^4F_{9/2}$ and ${}^4I_{13/2}$ level and a reduction of the population at the ${}^4S_{3/2}$ and ${}^4F_{7/2}$ level. Therefore, the red emission intensity is enhanced, while the green emission is extremely weak.

The EBT and CR processes significantly contribute to enhancing red emission through depopulating the ${}^2H_{11/2}$ and ${}^4S_{3/2}$ levels and increasing the population at the ${}^4F_{9/2}$ level [75]. The probability of energy transfer, as outlined by Dexter's energy transfer theory, is dependent on the distance $1/R^\theta$ between two ions. Consequently, the probability of $\text{CR}_{\text{Er-Er}}$ and $\text{EBT}_{\text{Er-Yb}}$ can be mathematically represented as:

$$P(\text{CR}) \propto 1/R_{\text{Er-Er}}^\theta \quad (4)$$

$$P(\text{EBT}) \propto 1/R_{\text{Er-Yb}}^\theta \quad (5)$$

where, $\mathbf{R}_{\text{Er-Er}}$ and $\mathbf{R}_{\text{Er-Yb}}$ represent the average distance between $\text{Er}^{3+} - \text{Er}^{3+}$ and $\text{Er}^{3+} - \text{Yb}^{3+}$, and θ represent a positive integer taking the values of 6, 8, and 10, which correspond to dipole-dipole, dipole-quadrupole, and quadrupole-quadrupole interactions, respectively. The $\mathbf{R}_{\text{Er-Er}}$ and $\mathbf{R}_{\text{Er-Yb}}$ can be expressed as:

$$R_{\text{Er-Er}} \approx 2 \left(\frac{3V}{4\pi C_{\text{Er}} N} \right)^{1/3} \quad (6)$$

$$R_{Er-Yb} \approx 2 \left(\frac{3V}{4\pi C_{Er+Yb}N} \right)^{1/3} \quad (7)$$

where V represents the unit cell volume, N is the number of substitutable sites for RE^{3+} in each unit cell, C_{Er} and C_{Er+Yb} are Er^{3+} doping concentration and total doping concentration of Yb^{3+} and Er^{3+} , respectively. For cubic Yb_2O_3 , the value of N is 32 and the unit cell volume is $1135.93 \text{ \AA}^3$. Furthermore, in the Yb_2O_3 matrix, Yb^{3+} ions not only contribute to the crystal structure, but also serve as sensitizers in the UC process. As a result, the total concentration of Yb^{3+} and Er^{3+} ions are fixed at 0.97, excluding the 3at% of ZrO_2 used as a sintering aid. Table 2 summarizes the ion distance of Yb_2O_3 : Er and other Yb based materials. It is obvious that the increasing Er^{3+} concentration will decrease the R_{Er-Er} and the probability of CR occurrence will increase according to the Eq. (4). Clearly, compared to $Yb_3Al_5O_{12}$ and $Yb_2Ti_2O_7$, Yb_2O_3 has a smaller Yb ion spacing because its cation composition is only Yb^{3+} . Additionally, the R_{Er-Yb} of Yb_2O_3 is considerably shorter than in Gd_2O_3 and Y_2O_3 hosts, which possess similar Ia3 structures. This suggests that a high concentration of Yb^{3+} would result in a reduction of the R_{Er-Yb} , and thus increase the probability of EBT as predicted by Eq. (5). The EBT process increases the population in $^4I_{13/2}$ level by decreasing the population of the $^4S_{3/2}$ level, and the ET3 process further populates the $^4F_{9/2}$ level [76]. The process described above could cause an increase in the population at the red-emitting $^4F_{9/2}$ level and a decrease at the green-emitting $^4S_{3/2}$ and $^2H_{11/2}$ level. Specifically, the strong red UC and negligible green UC.

Table 2. Volume of a unit cell (V), number of formula units in a unit cell (N), doping concentration of RE^{3+} (C_{Er-Er} and C_{Er-Yb}), and distance between RE^{3+} (R_{Er-Er} and R_{Er-Yb}) of Yb_2O_3 , Y_2O_3 , Gd_2O_3 and other Yb-based matrices.

Materials	$V(\text{\AA}^3)$	N	$C_{Er-Er}(\text{mol})$	$C_{Er+Yb}(\text{mol})$	$R_{Er-Er}(\text{\AA})$	$R_{Er-Yb}(\text{\AA})$
Yb_2O_3	1135.93	32	0.01	0.97	18.93	4.12
Yb_2O_3	1137.24	32	0.03	0.97	13.13	4.12
Yb_2O_3	1137.57	32	0.06	0.97	10.42	4.12
Yb_2O_3	1139.20	32	0.10	0.97	8.79	4.12
Yb_2O_3	1141.49	32	0.20	0.97	6.98	4.13
YbF_3 [75]	186.94	4	0.01	0.99	20.74	4.47
$Yb_3Al_5O_{12}$ [2]	1707.61	8	0.01	1	34.37	7.41
$Yb_2Ti_2O_7$ [44]	1028.16	16	0.01	1	23.07	4.97
Y_2O_3 [63]	1196.15	32	0.01	0.22	19.26	6.87
Gd_2O_3 [74]	1264.26	32	0.01	0.22	19.61	7.00

3.4. Optical thermometry behavior

The LIR technique is based on the temperature-dependent UCL of RE ions at two thermally-coupled energy levels, which makes it independent of excitation intensity fluctuations and spectral losses, resulting in high resolution and accuracy [77]. The autofluorescence of biological tissues can be minimized in an optical window ranging from the red region (600-700 nm) and the near-infrared region (700-1100 nm), thus it would be effective if pure red emission is studied as an application for optical thermometry. In most previous reports on optical thermometry based on LIR, the thermally-coupled $^2H_{11/2}$ and $^4S_{3/2}$ levels of Er^{3+} were the most commonly employed due to appropriate energy gap and the excellent sensing properties [78]. However, in this work, this pair of thermally-coupled levels could not be utilized for temperature sensing due to the extremely weak green UC emission. Fortunately, the red-emitting Stark sublevels of $^4F_{9/2}$ of Er^{3+} can also be employed to optical sensing with perfect sensitivity [44,76,79,80]. For example, Suo et al. reported on the use of two Stark sublevels of the $^4F_{9/2}$ manifold in Er^{3+} for

optical thermometry [81]. In $\text{Yb}_2\text{O}_3:\text{Er}$ system, the splitting in UCL spectrum originates from the splitting of the $^4\text{F}_{9/2}$ level into Stark sublevels, designated as $^4\text{F}_{9/2}(0)$, $^4\text{F}_{9/2}(1)$ and $^4\text{F}_{9/2}(2)$. To distinguish the transitions between Stark sub-levels of $^4\text{I}_{15/2}$ and $^4\text{F}_{9/2}$ states, Wu. et al. has measured the $^4\text{I}_{15/2}-^4\text{F}_{9/2}$ absorption spectrum and the $^4\text{F}_{9/2}-^4\text{I}_{15/2}$ emission spectrum at 77 K [73]. And the transitions from the populated $^4\text{F}_{9/2}$ level to the $^4\text{I}_{15/2}$ level give rise to a red emission from 648 to 685 nm with a split peak at 655, 662, and 685 nm. According to the peaks in the UCL spectrum measured at room temperature, the energy levels of $^4\text{F}_{9/2}$ in $\text{Yb}_2\text{O}_3:\text{Er}$ TCs were assigned as shown in Supplement 1, Fig. S10. Here, we focus on the 655 nm and 662 nm emission peaks because they are derived from the thermally coupled Stark level (2) and (1) of the $^4\text{F}_{9/2}$ manifold. Temperature-dependent UCL spectra of the $\text{Yb}_2\text{O}_3:\text{Er}$ TCs was shown in Fig. 4(a) and Supplement 1, Fig. S9. Apparently, the red emission peak did not shift and the red UCL intensity gradually decreased with the increase of temperature. For a more intuitive presentation, Fig. 4(b) displays bar graphs depicting the intensity changes at 662 nm and 655 nm with varying temperatures. In addition, Fig. 4(c) exhibits the temperature-dependent UC spectra, which have been normalized by the intensity at 662 nm. Certainly, as the temperature increases, the emission intensity at both 662 nm and 655 nm will decrease. However, the intensity at 662 nm experiences a more rapid decrease in comparison to that at 655 nm. The color coordinates in CIE chromaticity diagram of UCL (shown in Fig. 4(d)) slightly towards the green direction as the temperature rises, which may be attributed to the decrease of UCL intensity leads to the reduction of signal-to-noise ratio.

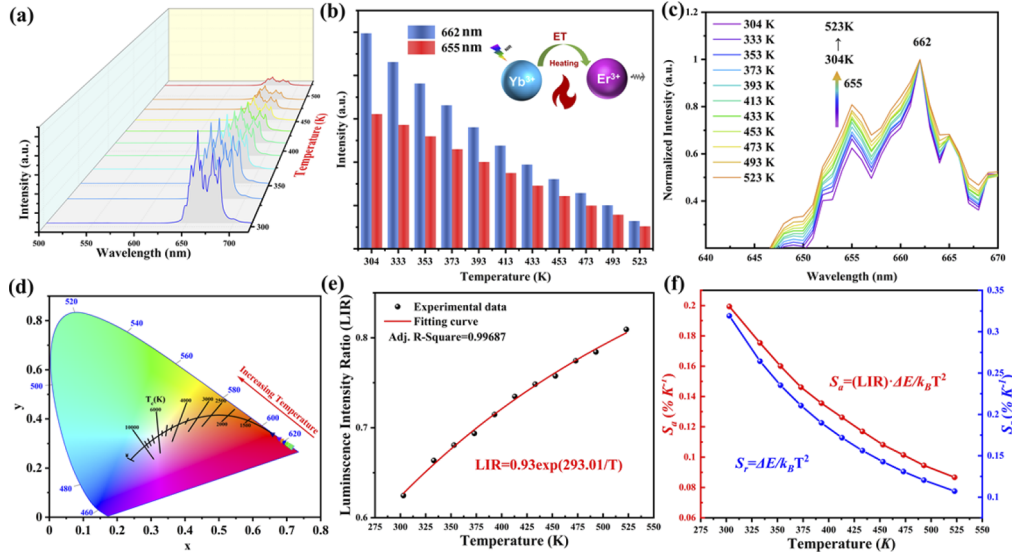


Fig. 4. (a) Temperature-dependence UC emission spectra in the range 304 K–523 K. (b) UC luminescence intensity at 662 nm and 655 nm for $\text{Yb}_2\text{O}_3: 6\% \text{Er}$. (c) Temperature-dependence UC fluorescence spectra normalized by emission intensity at 662 nm. (d) CIE diagram of 6% $\text{Yb}_2\text{O}_3: \text{Er}^{3+}$ TCs with increasing temperature. (e) Experimental and fitted I_{655}/I_{662} LIR versus temperature plots for the red UC emission. (f) Sensitivity variation with temperature obtained from the red UC emission.

The LIR for transition from two thermally-coupled levels could be expressed using the Boltzmann distribution law as [76]:

$$LIR = \frac{I_U}{I_L} = B \exp\left(-\frac{\Delta E}{k_B T}\right) \quad (8)$$

with $B = \frac{g_U \delta_U \omega_U}{g_L \delta_L \omega_L}$. I , δ , g and ω are the fluorescence intensity, radiative rate, degeneracy degree, and transitions angular frequency, respectively. The subscript U and L denote the thermally-coupled upper and lower level, respectively. T , ΔE , and k_B define the absolute temperature, energy separation, and Boltzmann constant, respectively. Based on Eq. 8, the variation of LIR between 655/662 nm with temperature could be expressed as: (shown in Fig. 4(e))

$$LIR = 0.93 \exp\left(\frac{293.01}{T}\right) \quad (9)$$

since $\Delta E/k_B = 293.01$, the calculated ΔE between peak 655 nm and peak 662 nm is approximately 159 cm^{-1} . Compared with the ΔE obtained from the UCL spectrum (203 cm^{-1}), the error of ΔE is 44 cm^{-1} , attributed to the enhancement of the multi-phonon relaxation and ET at higher temperature [82]. The absolute sensitivity S_a and relative sensitivity S_r , which are the important parameters to evaluate the temperature sensor, could be calculated using the following equation:

$$S_a = \left| \frac{dLIR}{dT} \right| = LIR \left(\frac{\Delta E}{k_B T^2} \right) \quad (10)$$

$$S_r = \left| \frac{dLIR}{dT} \frac{1}{LIR} \right| = \frac{\Delta E}{k_B T^2} \quad (11)$$

The sensitivity of the optical thermometry was calculated as a function of temperature and presented in Fig. 4(f). The values of both S_a and S_r keep decreasing in the testing range from 304 K to 523 K, which achieve the maximum $0.199\% \text{ K}^{-1}$ and $0.319\% \text{ K}^{-1}$ respectively at 304 K. Table 3 shows the comparison of temperature measurement sensitivity between Yb_2O_3 : Er TCs and previously reported $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped materials based on $^4\text{F}_{9/2}$ energy levels. The temperature sensitivity of the sensor is strongly influenced by the ΔE between the thermally coupled levels, so the sensitivity at the same thermally coupled energy level does not differ much in different matrices. This shows that this Yb_2O_3 : Er TCs with both observation and temperature detection properties, has good prospects for application in special scenarios.

Table 3. Optical thermometric parameters in Er^{3+} doped materials based on $^4\text{F}_{9/2}$ level using the LIR technique

Compounds	λ_{em} (nm)	Range (K)	S_a (K^{-1})	S_r ($\% \text{ K}^{-1}$)	Ref.
$\text{CaO-Y}_2\text{O}_3$: $\text{Er}^{3+}/\text{Yb}^{3+}$	648, 661	298–573	0.0006 (298 K)	$527/T^2$	[76]
$\text{Ba}_3\text{Y}_4\text{O}_9$: $\text{Er}^{3+}/\text{Yb}^{3+}$	656, 664	83–563	0.0037 (143 K)	$269.5/T^2$	[80]
$\text{Ba}_3\text{Y}_4\text{O}_9$: $\text{Er}^{3+}/\text{Yb}^{3+}$	664, 677	298–573	0.0016 (298 K)	$146.3/T^2$	[73]
$\text{Ba}_5\text{Gd}_8\text{Zn}_4\text{O}_{21}$: $\text{Er}^{3+}/\text{Yb}^{3+}$	653, 674	200–490	0.0029 (200 K)	$143.6/T^2$	[81]
$\text{Gd}_2(\text{WO}_4)_3$: $\text{Er}^{3+}/\text{Yb}^{3+}$	657, 666	296–620	0.0013 (296 K)	$110/T^2$	[79]
ZrO_2 : $\text{Er}^{3+}/\text{Yb}^{3+}/\text{Li}^+$	646, 678	323–673	0.0026 (110 K)	$220/T^2$	[83]
$\text{Ba}_2\text{TiGe}_2\text{O}_8$: $\text{Er}^{3+}/\text{Yb}^{3+}$	662, 681	333–573	0.0063 (333 K)	$295.63/T^2$	[84]
BaMgF_4 : $\text{Er}^{3+}/\text{Yb}^{3+}$	651, 668	323–593	0.0016 (323 K)	$139.19/T^2$	[85]
Yb_2O_3 : Er^{3+} TCs	655, 662	304–523	0.00199 (304 K)	$293.01/T^2$	This work

4. Conclusion

In this work, we fabricated a series of Er^{3+} doped highly transparent Yb_2O_3 ceramics with distinct Er^{3+} concentrations. The phase and structural evolution of ceramics was systematically determined. All samples exhibit high optical quality with in-line transmittance $>80\%$. In the 980nm-excited UC spectrum, green emission does not appear, thus Yb_2O_3 : Er TCs exhibit

ultra-pure single band red emission, with CIE coordinates of (0.72, 0.28) and color purity exceeding 99.9%. This is the first demonstration of pure red UC luminescence in TCs to the best of our knowledge. The UC mechanism was discussed through the pump power dependence of UCL intensity, which was determined to be a two-phonon process. The generation of pure red UCL is attributed to the EBT and CR processes, as these processes result in a decrease of the $^2H_{11/2}$ and $^4S_{3/2}$ green UC levels and an increase of the $^4F_{9/2}$ red UC level. The optical thermometry behaviors of this pure red emission based on the Stark sublevels of Er^{3+} $^4F_{9/2}$ have been investigated in detail. The maximum S_a and S_r were calculated to be $0.199\%K^{-1}$ and $0.319\%K^{-1}$, respectively. Combined with the strong penetration and anti-interference capabilities of red UCL with the desirable transparency, high thermal conductivity and thermal stability, and excellent mechanical properties of TC, $Yb_2O_3: Er$ TCs are extremely promising multi-functional optical materials.

Funding. National Natural Science Foundation of China (U21A20441).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See [Supplement 1](#) for supporting content.

References

1. F. Auzel, "Compteur quantique par transfert d'energie entre deux ions de terres rares dans un tungstate mixte et dans un verre," *Comptes Rendus Hebdomadaires Des Seances De L Academie Des Sciences Serie B* **262**, 1016 (1966).
2. B. Dong, B. Cao, Y. He, Z. Liu, Z. Li, and Z. Feng, "Temperature sensing and in vivo imaging by molybdenum sensitized visible upconversion luminescence of rare-earth oxides," *Adv. Mater.* **24**(15), 1987–1993 (2012).
3. D. Hudry, I. A. Howard, R. Popescu, D. Gerthsen, and B. S. Richards, "Structure–Property Relationships in Lanthanide-Doped Upconverting Nanocrystals: Recent Advances in Understanding Core–Shell Structures," *Adv. Mater.* **31**(26), 1900623 (2019).
4. B. S. Richards, D. Hudry, D. Busko, A. Turshatov, and I. A. Howard, "Photon Upconversion for Photovoltaics and Photocatalysis: A Critical Review," *Chem. Rev.* **121**(15), 9165–9195 (2021).
5. R. Deng, F. Qin, R. Chen, W. Huang, M. Hong, and X. Liu, "Temporal full-colour tuning through non-steady-state upconversion," *Nat. Nanotechnol.* **10**(3), 237–242 (2015).
6. A. Boccolini, J. Marques-Hueso, D. Chen, Y. Wang, and B. S. Richards, "Physical performance limitations of luminescent down-conversion layers for photovoltaic applications," *Sol. Energy Mater. Sol. Cells* **122**, 8–14 (2014).
7. G. Gao, D. Busko, S. Kauffmann-Weiss, A. Turshatov, I. A. Howard, and B. S. Richards, "Wide-range non-contact fluorescence intensity ratio thermometer based on Yb^{3+}/Nd^{3+} co-doped La_2O_3 microcrystals operating from 290 to 1230 K," *J. Mater. Chem. C* **6**(15), 4163–4170 (2018).
8. D. H. Kim, A. D'Aléo, X. K. Chen, A. D. S. Sandanayaka, D. Yao, L. Zhao, T. Komino, E. Zaborova, G. Canard, Y. Tsuchiya, E. Choi, J. W. Wu, F. Fages, J. L. Brédas, J. C. Ribierre, and C. Adachi, "High-efficiency electroluminescence and amplified spontaneous emission from a thermally activated delayed fluorescent near-infrared emitter," *Nat. Photonics* **12**(2), 98–104 (2018).
9. S. Chen, A. Z. Weitemier, X. Zeng, L. He, X. Wang, Y. Tao, A. J. Y. Huang, Y. Hashimoto-dani, M. Kano, H. Iwasaki, L. K. Parajuli, S. Okabe, D. B. Loong Teh, A. H. All, I. Tsutsui-Kimura, K. F. Tanaka, X. Liu, and T. J. McHugh, "Near-infrared deep brain stimulation via upconversion nanoparticle-mediated optogenetics," *Science* **359**(6376), 679–684 (2018).
10. J. Wang, R. Deng, M. A. MacDonald, B. Chen, J. Yuan, F. Wang, D. Chi, T. S. Andy Hor, P. Zhang, G. Liu, Y. Han, and X. Liu, "Enhancing multiphoton upconversion through energy clustering at sublattice level," *Nat. Mater.* **13**(2), 157–162 (2014).
11. J. M. Meruga, A. Baride, W. Cross, J. J. Kellar, and P. S. May, "Red-green-blue printing using luminescence-upconversion inks," *J. Mater. Chem. C* **2**(12), 2221–2227 (2014).
12. L. Zhou, R. Wang, C. Yao, X. Li, C. Wang, X. Zhang, C. Xu, A. Zeng, D. Zhao, and F. Zhang, "Single-band upconversion nanoprobes for multiplexed simultaneous in situ molecular mapping of cancer biomarkers," *Nat. Commun.* **6**(1), 6938 (2015).
13. H. Huang, Z. Wang, B. Huang, P. Wang, X. Zhang, X. Qin, Y. Dai, G. Zhou, and M. H. Whangbo, "Intense Single Red Emission Induced by Near-Infrared Irradiation Using a Narrow Bandgap Oxide $BiVO_4$ as the Host for Yb^{3+} and Tm^{3+} Ions," *Adv. Opt. Mater.* **6**(15), 1701331 (2018).
14. B. S. Moon, H. E. Kim, and D. H. Kim, "Ultrafast Single-Band Upconversion Luminescence in a Liquid-Quenched Amorphous Matrix," *Adv. Mater.* **30**(25), 1800008 (2018).
15. X. F. Yu, L. D. Chen, M. Li, M. Y. Xie, L. Zhou, Y. Li, and Q. Q. Wang, "Highly efficient fluorescence of NdF_3/SiO_2 core/shell nanoparticles and the applications for in vivo NIR detection," *Adv. Mater.* **20**(21), 4118–4123 (2008).

16. M. Ding, S. Cui, L. Fang, Z. Lin, C. Lu, and X. Yang, "NIR-I-Responsive Single-Band Upconversion Emission through Energy Migration in Core-Shell-Shell Nanostructures," *Angew. Chem. Int. Ed.* **61**(29), e202203631 (2022).
17. D. Su, X. Zhao, X. Yan, X. Han, Z. Zhu, C. Wang, X. Jia, F. Liu, P. Sun, X. Liu, and G. Lu, "Background-free sensing platform for on-site detection of carbamate pesticide through upconversion nanoparticles-based hydrogel suit," *Biosens. Bioelectron.* **194**, 113598 (2021).
18. D. Yang, P. Ma, Z. Hou, Z. Cheng, C. Li, and J. Lin, "Current advances in lanthanide ion (Ln³⁺)-based upconversion nanomaterials for drug delivery," *Chem. Soc. Rev.* **44**(6), 1416–1448 (2015).
19. G. Chen, H. Qiu, P. N. Prasad, and X. Chen, "Upconversion nanoparticles: Design, nanochemistry, and applications in Theranostics," *Chem. Rev.* **114**(10), 5161–5214 (2014).
20. G. Tian, Z. Gu, L. Zhou, W. Yin, X. Liu, L. Yan, S. Jin, W. Ren, G. Xing, S. Li, and Y. Zhao, "Mn²⁺ + dopant-controlled synthesis of NaYF₄:Yb/Er upconversion nanoparticles for in vivo imaging and drug delivery," *Adv. Mater.* **24**(9), 1226–1231 (2012).
21. M. Ding, M. Xu, C. Lu, J. Xi, Z. Ji, and D. Chen, "808 nm NIR light excited single-band red upconversion emission in lanthanide-doped KMnF₃ nanocrystals," *J. Alloys Compd.* **721**, 531–537 (2017).
22. Z. Bai, H. Lin, J. Johnson, S. C. Rong Gui, K. Imakita, R. Montazami, M. Fujii, and N. Hashemi, "The single-band red upconversion luminescence from morphology and size controllable Er³⁺/Yb³⁺ + doped MnF₂ nanostructures," *J. Mater. Chem. C* **2**(9), 1736–1741 (2014).
23. Y. Zhang, J. D. Lin, V. Vijayaragavan, K. K. Bhakoo, and T. Thatt Yang Tan, "Tuning sub-10 nm single-phase NaMnF₃ nanocrystals as ultrasensitive hosts for pure intense fluorescence and excellent T1 magnetic resonance imaging," *Chem. Commun.* **48**(83), 10322–10324 (2012).
24. M. Tuomisto, E. Palo, T. Laihin, I. Hyppänen, M. Lastusaari, H. C. Swart, and J. Hölsä, "Effect of Mn and Cr doping on the up-conversion luminescence from NaYF₄:Yb³⁺,Er³⁺," *Opt. Mater.* **59**, 115–119 (2016).
25. B. B. Srivastava, S. K. Gupta, and Y. Mao, "Single red emission from upconverting ZnGa₂O₄:Yb,Er nanoparticles co-doped by Cr³⁺," *J. Mater. Chem. C* **8**(19), 6370–6379 (2020).
26. J. Wang, F. Wang, C. Wang, Z. Liu, and X. Liu, "Single-band upconversion emission in lanthanide-doped KMnF₃ nanocrystals," *Angew. Chem. Int. Ed.* **50**(44), 10369–10372 (2011).
27. X. Teng, Y. Zhu, W. Wei, S. Wang, J. Huang, R. Naccache, W. Hu, A. I. Y. Tok, Y. Han, Q. Zhang, Q. Fan, W. Huang, J. A. Capobianco, and L. Huang, "Lanthanide-doped Na xScF₃ + x nanocrystals: Crystal structure evolution and multicolor tuning," *J. Am. Chem. Soc.* **134**(20), 8340–8343 (2012).
28. D. Chen, L. Lei, R. Zhang, A. Yang, J. Xu, and Y. Wang, "Intrinsic single-band upconversion emission in colloidal Yb/Er(Tm): Na₃Zr(Hf)F₇ nanocrystals," *Chem. Commun.* **48**(86), 10630–10632 (2012).
29. Y. Wang, T. Wen, H. Zhang, J. Sun, M. Zhang, Y. Guo, W. Luo, M. Xia, Y. Wang, and B. Yang, "Low-temperature fluorination route to lanthanide-doped monoclinic ScOF host material for tunable and nearly single band up-conversion luminescence," *J. Phys. Chem. C* **118**(19), 10314–10320 (2014).
30. M. Wu, E. H. Song, Z. T. Chen, S. Ding, S. Ye, J. J. Zhou, S. Q. Xu, and Q. Y. Zhang, "Single-band red upconversion luminescence of Yb³⁺+Er³⁺ + via nonequivalent substitution in perovskite KMgF₃ nanocrystals," *J. Mater. Chem. C* **4**(8), 1675–1684 (2016).
31. Y. Ding, X. Teng, H. Zhu, L. Wang, W. Pei, J. J. Zhu, L. Huang, and W. Huang, "Orthorhombic KSc₂F₇:Yb/Er nanorods: Controlled synthesis and strong red upconversion emission," *Nanoscale* **5**(23), 11928–11932 (2013).
32. H. Dong, L. D. Sun, Y. F. Wang, J. Ke, R. Si, J. W. Xiao, G. M. Lyu, S. Shi, and C. H. Yan, "Efficient Tailoring of Upconversion Selectivity by Engineering Local Structure of Lanthanides in Na_xREF₃ + x Nanocrystals," *J. Am. Chem. Soc.* **137**(20), 6569–6576 (2015).
33. W. Xu, L. Lei, Y. Wang, E. Liu, L. Chen, and S. Xu, "Modulating electron population pathways for time-dependent dynamic multicolor displays," *Mater. Horiz.* **8**(12), 3443–3448 (2021).
34. X. Zheng, R. K. Kankala, C. G. Liu, Y. Wen, S. B. Wang, A. Z. Chen, and Y. Zhang, "Tailoring Lanthanide Upconversion Luminescence through Material Designs and Regulation Strategies," *Adv. Opt. Mater.* **10**(12), 2200167 (2022).
35. W. Wei, Y. Zhang, R. Chen, J. Goggi, N. Ren, L. Huang, K. K. Bhakoo, H. Sun, and T. T. Y. Tan, "Cross Relaxation Induced Pure Red Upconversion in Activator- and Sensitizer-Rich Lanthanide Nanoparticles," *Chem. Mater.* **26**(18), 5183–5186 (2014).
36. C. Xu, Q. Yang, G. Ren, and Y. Liu, "Pure red upconversion emission from Yb₃Al₅O₁₂ phase doped with high Er³⁺ + concentration," *J. Alloys Compd.* **503**(1), 82–85 (2010).
37. T. Wen, Y. Zhou, Y. Guo, C. Zhao, B. Yang, and Y. Wang, "Color-tunable and single-band red upconversion luminescence from rare-earth doped Vernier phase ytterbium oxyfluoride nanoparticles," *J. Mater. Chem. C* **4**(4), 684–690 (2016).
38. N. Zhang, M. S. Molokeev, Q. Liu, and Z. Xia, "Pure red upconversion luminescence and optical thermometry of Er³⁺ + doped sensitizer-rich SrYbInO₄ phosphors," *J. Mater. Chem. C* **6**(27), 7361–7366 (2018).
39. M. Shen, Z. Cheng, Y. Li, T. Liu, Y. Peng, and Y. Zhang, "The single-band red upconversion emission of Er³⁺ + in YbOCl layered structure," *J. Mater. Sci.: Mater. Electron.* **32**, 7026–7033 (2021).
40. S. Wen, J. Zhou, K. Zheng, A. Bednarkiewicz, X. Liu, and D. Jin, "Advances in highly doped upconversion nanoparticles," *Nat. Commun.* **9**(1), 2415 (2018).

41. X. Xie, Q. Li, H. Chen, W. Wang, F. Wu, L. Tu, Y. Zhang, X. Kong, and Y. Chang, "Manipulating the Injected Energy Flux via Host-Sensitized Nanostructure for Improving Multiphoton Upconversion Luminescence of Tm³⁺," *Nano Lett.* **22**(13), 5339–5347 (2022).
42. J. Hu, D. Guan, B. Zhao, Y. Zhang, X. Qiu, Y. Zhang, and Q. Liu, "Ytterbium-Enriched Outmost Shell for Enhanced Upconversion Single Molecule Imaging and Interfacial Triplet Energy Transfer," *Adv. Opt. Mater.* **10**(6), 2101763 (2022).
43. K. Lu, Y. Ye, W. Han, C. Zhang, L. Liang, H. Yuan, Y. Shi, J. Qi, and T. Lu, "Defect Regulation of Terbium-Doped Y₂Zr₂O₇ Transparent Ceramic for Wide-Range Multimode Tunable Light-Shielding Windows," *J. Phys. Chem. Lett.* **13**(34), 8097–8103 (2022).
44. Y. Ye, K. Lu, and J. Qi, "Developing Smart Temperature Sensing Window Based on Highly Transparent Rare-Earth Doped Yttrium Zirconate Ceramics," *ACS Appl. Mater. Interfaces* **14**(34), 39072–39080 (2022).
45. F. Yu, B. Ma, Z. Xie, P. Wang, X. Wu, C. Lin, C. Zhao, M. Gao, L. Luo, and Q. Zhang, "Developing "smart windows" for optical storage and temperature sensing based on KNN transparent ceramics," *J. Eur. Ceram. Soc.* **43**(10), 4408–4418 (2023).
46. W. Xu, X. Gao, L. Zheng, Z. Zhang, and W. Cao, "An optical temperature sensor based on the upconversion luminescence from Tm³⁺/Yb³⁺ + codoped oxyfluoride glass ceramic," *Sens. Actuators, B* **173**, 250–253 (2012).
47. Q. Du, J. Ueda, R. Zheng, and S. Tanabe, "Photochromism and Long Persistent Luminescence in Pr³⁺-Doped Garnet Transparent Ceramic via UV or Blue Light Up-Conversion Charging," *Adv. Opt. Mater.* **11**(7), 2202612 (2023).
48. T. Ji, T. Wang, H. Li, Q. Peng, H. Tang, S. Hu, A. Yakovlev, Y. Zhong, and X. Xu, "Ce³⁺-Doped Yttrium Aluminum Garnet Transparent Ceramics for High-Resolution X-Ray Imaging," *Adv. Opt. Mater.* **10**(6), 2102056 (2022).
49. K. Al Saghir, S. Chenu, E. Veron, F. Fayon, M. Suchomel, C. Genevois, F. Porcher, G. Matzen, D. Massiot, and M. Allix, "Transparency through structural disorder: A new concept for innovative transparent ceramics," *Chem. Mater.* **27**(2), 508–514 (2015).
50. M. Sakajo, V. Beilin, M. Mann-Lahav, S. Zamir, G. E. Shter, and G. S. Grader, "Highly Transparent Polycrystalline MgO via Spark Plasma Sintering," *ACS Appl. Mater. Interfaces* **14**(46), 52108–52116 (2022).
51. J. Lin, P. Wang, H. Wang, Y. Shi, K. Zhu, F. Yan, G. Li, H. Ye, J. Zhai, and X. Wu, "Significantly Photo-Thermochromic KNN-Based "Smart Window" for Sustainable Optical Data Storage and Anti-Counterfeiting," *Adv. Opt. Mater.* **9**(17), 2100580 (2021).
52. A. Ikesue, T. Kinoshita, K. Kamata, and K. Yoshida, "Fabrication and Optical Properties of High-Performance Polycrystalline Nd:YAG Ceramics for Solid-State Lasers," *J Am Ceram Soc* **78**(4), 1033–1040 (1995).
53. B. Sun, L. Zhang, L. Zhang, T. Zhou, C. Shao, Y. Ma, Q. Yao, Z. Jiang, F. A. Selim, and H. Chen, "Protected-annealing regulated defects to improve optical properties and luminescence performance of Ce:YAG transparent ceramics for white LEDs," *J. Mater. Chem. C* **7**(14), 4057–4065 (2019).
54. H. Ding, Z. Liu, P. Hu, Y. Liu, P. Sun, Z. Luo, X. Chen, H. Jiang, and J. Jiang, "High Efficiency Green-Emitting LuAG:Ce Ceramic Phosphors for Laser Diode Lighting," *Adv. Opt. Mater.* **9**(8), 2002141 (2021).
55. C. Zhang, X. Wang, L. Liang, H. Yuan, J. Qi, Z. Liao, and T. Lu, "Vacuum sintering of Yb₂O₃ transparent ceramics: Effect of ZrO₂ concentration on structural and optical properties," *J. Alloys Compd.* **907**, 164454 (2022).
56. Z. Huang, Y. Shi, Y. Zhang, M. Deng, Y. Guo, Q. Kong, J. Qi, B. Ma, Q. Wang, and H. Wang, "An effective strategy for preparing transparent ceramics using nanorod powders based on pressure-assisted particle fracture and rearrangement," *J. Adv. Ceram.* **11**(12), 1976–1987 (2022).
57. M. I. Mendelson, "Average Grain Size in Polycrystalline Ceramics," *J Am Ceram Soc* **52**(8), 443–446 (1969).
58. H. Wu, Z. Hao, L. Zhang, X. Zhang, Y. Xiao, G.-H. Pan, H. Wu, Y. Luo, H. Zhao, and J. Zhang, "Phonon Energy Dependent Energy Transfer Upconversion for the Red Emission in the Er³⁺/Yb³⁺ System," *J. Phys. Chem. C* **122**(17), 9611–9618 (2018).
59. T. Yanagida, K. Kamada, Y. Fujimoto, H. Yagi, and T. Yanagitani, "Comparative study of ceramic and single crystal Ce:GAGG scintillator," *Opt. Mater.* **35**(12), 2480–2485 (2013).
60. J. Zhou, Q. Liu, W. Feng, Y. Sun, and F. Li, "Upconversion Luminescent Materials: Advances and Applications," *Chem. Rev.* **115**(1), 395–465 (2015).
61. C. M. S. Jones, N. Panov, A. Skripka, J. Gibbons, F. Hesse, J.-W. G. Bos, X. Wang, F. Vetrone, G. Chen, E. Hemmer, and J. Marques-Hueso, "Effect of light scattering on upconversion photoluminescence quantum yield in microscale-to-nanoscale materials," *Opt. Express* **28**(15), 22803–22818 (2020).
62. K. Feng, B. Lu, and L. Hu, "Down/Upconversion Luminescence Behaviors and Temperature-Sensing Properties of Highly Transparent (Er_{1-x}Yb_x)₂O₃ Ceramics," *ACS Appl. Electron. Mater.* **4**(2), 761–767 (2022).
63. J. Zhou, Y. Chen, R. Lei, H. Wang, and S. Xu, "Role of Er³⁺ + concentration on the photoluminescence and temperature sensing properties in Er³⁺ + doped Y₂O₃ based transparent ceramics," *J. Lumin.* **215**, 116671 (2019).
64. A. Boccolini, E. Favilla, M. Tonelli, B. S. Richards, and R. R. Thomson, "Highly efficient upconversion in Er³⁺ + doped BaY₂F₈ single crystals: dependence of quantum yield on excitation wavelength and thickness," *Opt. Express* **23**(15), A903–A915 (2015).
65. A. Boccolini, J. Marques-Hueso, and B. S. Richards, "Self-absorption in upconverter luminescent layers: impact on quantum yield measurements and on designing optimized photovoltaic devices," *Opt. Lett.* **39**(10), 2904–2907 (2014).

66. R. Joshi, R. S. Perala, S. B. Shelar, A. Ballal, B. P. Singh, and R. S. Ningthoujam, "Super bright red upconversion in NaErF₄:0.5%Tm@NaYF₄:20%Yb nanoparticles for anti-counterfeit and bioimaging applications," *ACS Appl. Mater. Interfaces* **13**(2), 3481–3490 (2021).
67. W. Zheng, B. Sun, Y. Li, T. Lei, R. Wang, and J. Wu, "Low Power High Purity Red Upconversion Emission and Multiple Temperature Sensing Behaviors in Yb³⁺,Er³⁺ Codoped Gd₂O₃ Porous Nanorods," *ACS Sustainable Chem. Eng.* **8**(25), 9578–9588 (2020).
68. F. Hu, J. Cao, X. Wei, X. Li, J. Cai, H. Guo, Y. Chen, C. K. Duan, and M. Yin, "Luminescence properties of Er³⁺-doped transparent NaYb₂F₇ glass-ceramics for optical thermometry and spectral conversion," *J. Mater. Chem. C* **4**(42), 9976–9985 (2016).
69. M. Pollnau, D. R. Gamelin, S. R. Lüthi, H. U. Güdel, and M. P. Hehlen, "Power dependence of upconversion luminescence in lanthanide and transition-metal-ion systems," *Phys. Rev. B* **61**(5), 3337–3346 (2000).
70. Y. Lei, H. Song, L. Yang, L. Yu, Z. Liu, G. Pan, X. Bai, and L. Fan, "Upconversion luminescence, intensity saturation effect, and thermal effect in Gd₂O₃:Er³⁺,Yb³⁺ + nanowires," *J Chem Phys* **123**(17), 174710 (2005).
71. W. Guo, H. Hao, S. Jin, Q. Su, H. Li, X. Hu, Y. Gan, L. Qin, W. Gao, and G. Liu, "Preparation and infrared to visible upconversion luminescence of Yb₂O₃:Ho³⁺ + nanocrystalline powders," *Ceram. Int.* **43**(5), 4330–4334 (2017).
72. C. Lee, H. Park, W. Kim, and S. Park, "Origin of strong red emission in Er(3+)-based upconversion materials: role of intermediate states and cross relaxation," *Phys. Chem. Chem. Phys.* **21**(43), 24026–24033 (2019).
73. H. Wu, Z. Hao, L. Zhang, X. Zhang, Y. Xiao, G.-H. Pan, H. Wu, Y. Luo, L. Zhang, and J. Zhang, "Er³⁺/Yb³⁺ + codoped phosphor Ba₃Y₄O₉ with intense red upconversion emission and optical temperature sensing behavior," *J. Mater. Chem. C* **6**(13), 3459–3467 (2018).
74. J. Liu, H. Deng, Z. Huang, Y. Zhang, D. Chen, and Y. Shao, "Phonon-assisted energy back transfer-induced multicolor upconversion emission of Gd₂O₃:Yb³⁺/Er³⁺ + nanoparticles under near-infrared excitation," *Phys. Chem. Chem. Phys.* **17**(23), 15412–15418 (2015).
75. Z. Sun, M. Jia, Z. Fu, M. Zhang, H. Wang, and Y. Xu, "High-performance disease diagnosis fluorescent probe based on new type structure YbF₃: Er³⁺@SiO₂@GQDs," *Chem. Eng. J.* **406**, 126755 (2021).
76. G. Xiang, X. Liu, W. Liu, B. Wang, Z. Liu, S. Jiang, X. Zhou, L. Li, Y. Jin, and J. Zhang, "Multifunctional optical thermometry based on the stark sublevels of Er³⁺ + in CaO-Y₂O₃: Yb³⁺/Er³⁺," *J Am Ceram Soc* **103**(4), 2540–2547 (2020).
77. G. Gao, D. Busko, N. Katumo, R. Joseph, E. Madirov, A. Turshatov, I. A. Howard, and B. S. Richards, "Ratiometric Luminescent Thermometry with Excellent Sensitivity over a Broad Temperature Range Utilizing Thermally-Assisted and Multiphoton Upconversion in Triply-Doped La₂O₃:Yb³⁺/Er³⁺/Nd³⁺," *Adv. Opt. Mater.* **9**(5), 2001901 (2021).
78. C. Wang, Y. Jin, R. Zhang, Q. Yao, and Y. Hu, "A review and outlook of ratiometric optical thermometer based on thermally coupled levels and non-thermally coupled levels," *J. Alloys Compd.* **894**, 162494 (2022).
79. H. Lu, R. Meng, H. Hao, Y. Bai, Y. Gao, Y. Song, Y. Wang, and X. Zhang, "Stark sublevels of Er³⁺-Yb³⁺ + codoped Gd₂(WO₄)₃ phosphor for enhancing the sensitivity of a luminescent thermometer," *RSC Adv.* **6**(62), 57667–57671 (2016).
80. S. Liu, H. Ming, J. Cui, S. Liu, W. You, X. Ye, Y. Yang, H. Nie, and R. Wang, "Color-Tunable Upconversion Luminescence and Multiple Temperature Sensing and Optical Heating Properties of Ba₃Y₄O₉:Er³⁺/Yb³⁺ Phosphors," *J. Phys. Chem. C* **122**(28), 16289–16303 (2018).
81. H. Suo, C. Guo, and T. Li, "Broad-Scope Thermometry Based on Dual-Color Modulation up-Conversion Phosphor Ba₅Gd₈Zn₄O₂₁:Er³⁺/Yb³⁺," *J. Phys. Chem. C* **120**(5), 2914–2924 (2016).
82. C. Mi, J. Wu, Y. Yang, B. Han, and J. Wei, "Efficient upconversion luminescence from Ba₅Gd₈Zn₄O₂₁:Yb(3+), Er(3+) based on a demonstrated cross-relaxation process," *Sci. Rep.* **6**(1), 22545 (2016).
83. L. Liu, Y. Wang, X. Zhang, K. Yang, Y. Bai, C. Huang, and Y. Song, "Optical thermometry through green and red upconversion emissions in Er³⁺/Yb³⁺/Li⁺:ZrO₂ nanocrystals," *Opt. Commun.* **284**(7), 1876–1879 (2011).
84. B. Hou, M. Jia, P. Li, G. Liu, Z. Sun, and Z. Fu, "Multifunctional Optical Thermometry Based on the Rare-Earth-Ions-Doped Up-/Down-Conversion Ba₂TiGe₂O₈:Ln (Ln = Eu³⁺/ Er³⁺/ Ho³⁺/ Yb³⁺) Phosphors," *Inorg. Chem.* **58**(12), 7939–7946 (2019).
85. B. P. Kore, A. Kumar, L. Erasmus, R. E. Kroon, J. J. Terblans, S. J. Dhoble, and H. C. Swart, "Energy Transfer Mechanisms and Optical Thermometry of BaMgF₄:Yb³⁺,Er³⁺ Phosphor," *Inorg. Chem.* **57**(1), 288–299 (2018).
86. Y. Liu, C. Guo, G. Pan, J. Zhao, Z. Zhang, H. Gao, H. Zhang, J. Wei, and Y. Mao, "Highly efficient upconversion single red emission of hollow cubic α -NaErF₄ nanoparticles by Mn/Yb heavy doping," *J. Lumin.* **228**, 117637 (2020).